



4.5 Discrete random variables and the binomial distribution

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A discrete random variable X takes a countable set of values, each with probability $P(X=x)$, where $\sum P(X=x)=1$. Its expected value $E(X)=\sum x \cdot P(X=x)$ and variance $\text{Var}(X)=E(X^2)-[E(X)]^2$. The binomial distribution $X \sim B(n,p)$ models the number of successes in n independent trials each with success probability p : $P(X=x) = {}^nC_x \cdot p^x \cdot (1-p)^{(n-x)}$, with $E(X)=np$ and $\text{Var}(X)=np(1-p)$.

Worked example: follow the reasoning

$X \sim B(8, 0.3)$. Find $P(X=3)$ and $E(X)$.

$$P(X=3) = {}^8C_3 (0.3)^3(0.7)^5 = 56 \times 0.027 \times 0.16807 = 56 \times 0.004538 = 0.254 \text{ (3 s.f.)}$$

$$E(X) = np = 8(0.3) = 2.4.$$

Quick check · multiple choice

$X \sim B(10, 0.4)$. What is $\text{Var}(X)$?

- A 4
- B 2.4
- C 6
- D 1.55

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- A discrete random variable X has probability distribution given by $P(X=0)=0.2$, $P(X=1)=k$, $P(X=2)=0.35$, $P(X=3)=0.15$. [8]
 - Find the value of k .
 - Find $E(X)$.
 - A binomial variable $Y \sim B(20, 0.25)$ models a separate scenario. Find $P(Y=5)$ and $E(Y)$.

Final answer: _____



Answers and explanation

Question 1 [8]

(a) Sum to 1: $0.2+k+0.35+0.15=1 \rightarrow 0.7+k=1 \rightarrow k=0.3$.

(b) $E(X)=0(0.2)+1(0.3)+2(0.35)+3(0.15) = 0+0.3+0.7+0.45 = 1.45$.

(c) $P(Y=5)=20C5(0.25)^5(0.75)^{15} = 15504 \times 0.0009766 \times 0.01336 \approx 15504 \times 0.00001305 \approx 0.2023$. $E(Y)=np=20(0.25)=5$.

Quick check answer

B. $\text{Var}(X)=np(1-p)=10(0.4)(0.6)=2.4$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.